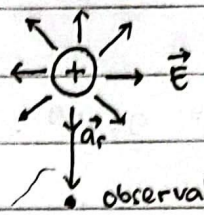
Medan Listrik

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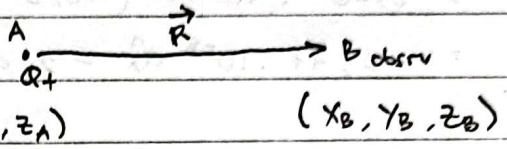
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Video 1 :

Medan Listrik : daerah pengaruh
: sumber listrik (muatan positif)



$$\vec{E} = k \frac{Q}{R^2} \vec{a}_R \left[\frac{\text{Volt}}{\text{meter}}\right]$$



$$\vec{R} = (x_B - x_A)\vec{a_x} + (y_B - y_A)\vec{a_y} + (z_B - z_A)\vec{a_z}$$

Video 2 :

$$\vec{E} = 9.10^9 \frac{Q}{\left(\sqrt{(x_B-x_A)^2 + (y_B-y_A)^2 + (z_B-z_A)^2}\right)^2} \frac{(x_B-x_A)\vec{a_x} + (y_B-y_A)\vec{a_y} + (z_B-z_A)\vec{a_z}}{\sqrt{(x_B-x_A)^2 + (y_B-y_A)^2 + (z_B-z_A)^2}}$$

$$\vec{E} = 9.10^9 \frac{Q}{\left[(x_B-x_A)^2 + (y_B-y_A)^2 + (z_B-z_A)^2\right]^{3/2}} (x_B-x_A)\vec{a_x} + (y_B-y_A)\vec{a_y} + (z_B-z_A)\vec{a_z}$$

ini kasus $\vec{E}$ di ruang hampa.

Jika di dielektrik, maka $k = \frac{9.10^9}{\varepsilon_r}$

Video 3 :

① $Q = 2\,nC$

S (2, 1, 0)

P

T (0, −1, 2)

$\vec{a_R}$

$\vec{E}$

$$\vec{E} = k \cdot \frac{Q}{R^2} \vec{a_R} = 9.10^9 \cdot \frac{Q}{R^2} \vec{a_R}$$

$$\vec{R}_{ST} = (0-2)\vec{a_x} + (-1-1)\vec{a_y} + (2-0)\vec{a_z}$$

$$= -2\vec{a_x} - 2\vec{a_y} + 2\vec{a_z}$$

jarak $R = |\vec{R}| = \sqrt{(-2)^2 + (-2)^2 + 2^2} = \sqrt{12} = 2\sqrt{3}$

$$\vec{a_R} = \frac{\vec{R}}{|\vec{R}|} = \frac{-2\vec{a_x} - 2\vec{a_y} + 2\vec{a_z}}{2\sqrt{3}} = \frac{-\vec{a_x} - \vec{a_y} + \vec{a_z}}{\sqrt{3}}$$

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$$\vec{E} = 9.10^9 \cdot \frac{2.10^{-9}}{(2\sqrt{3})^2} \cdot \left(\frac{-\hat{a}_x - \hat{a}_y + \hat{a}_z}{\sqrt{3}}\right)$$

$$= \frac{9.2}{4.3}\left(\frac{-\hat{a}_x - \hat{a}_y + \hat{a}_z}{\sqrt{3}}\right)$$

$$= \frac{(\sqrt{3})^4}{2(\sqrt{3})^3}\left(-\hat{a}_x - \hat{a}_y + \hat{a}_z\right) = -0{,}866\,\hat{a}_x - 0{,}866\,\hat{a}_y + 0{,}866\,\hat{a}_z \ N$$

Intensitas Medan Listrik nya $|\vec{E}| = \sqrt{(-0{,}86)^2 + (-0{,}86)^2 + (0{,}86)^2}$

$|\vec{E}| = 1{,}498$

Video 4 : jawaban video 3 ↑

Video 5 :

② jika $Q = -2nC$

$$\vec{E} = 9.10^9 \ \frac{(-2.10^{-9})}{(2\sqrt{3})^2}\left(\frac{-\hat{a}_x - \hat{a}_y + \hat{a}_z}{\sqrt{3}}\right)$$

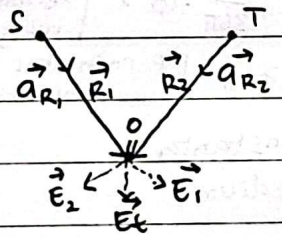
$$= \frac{9.2}{4.3}\left(\frac{\hat{a}_x + \hat{a}_y - \hat{a}_z}{\sqrt{3}}\right)$$

$$= \frac{\sqrt{3}}{2}\hat{a}_x + \frac{\sqrt{3}}{2}\hat{a}_y - \frac{\sqrt{3}}{2}\hat{a}_z \quad \left[\frac{\text{volt}}{\text{meter}}\right]$$

$|\vec{E}| = 1{,}498$

Video 6 ; Video 7 ; Video 8 ; Video 9

③



$Q_1 = 2nC$; $S(2, 1, 0)$

$Q_2 = 9nC$; $T(0, -1, 0)$

$O = (0, 0, 0)$

$\vec{E}_t = \vec{E}_1 + \vec{E}_2$

$\vec{R}_1 = (0-2)\hat{a}_x + (0-1)\hat{a}_y + (0-0)\hat{a}_z = -2\hat{a}_x - \hat{a}_y$

$|\vec{R}_1| = \sqrt{(-2)^2 + (-1)^2} = \sqrt{5}$

$\vec{a}_{R_1} = \dfrac{-2\vec{a}_x - \vec{a}_y}{\sqrt{5}}$

$\vec{R}_2 = (0-0)\vec{a}_x + (0-(-1))\vec{a}_y + (0-0)\vec{a}_z = \vec{a}_y$

$|\vec{R}_2| = 1 \Rightarrow \vec{a}_{R_2} = \vec{a}_y$

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$\vec{E_t} = \vec{E_1} + \vec{E_2}$

$= k \frac{Q_1}{R_1^2} \vec{a_{R_1}} + k \frac{Q_2}{R_2^2} \vec{a_{R_2}}$

$= 9.10^9 \left[\frac{2.10^{-9}}{(\sqrt{5})^2} \left(\frac{-2\vec{a_x} - \vec{a_y}}{\sqrt{5}} \right) + \frac{9.10^{-9}}{1^2} (\vec{a_y}) \right]$

$= 9.10^9 \left[-0,36\ \vec{a_x} - 0,179\ \vec{a_y} + 9\ \vec{a_y} \right] .10^{-9}$

$= 9 \left[-0,36\ \vec{a_x} + 8,82\ \vec{a_y} \right]$

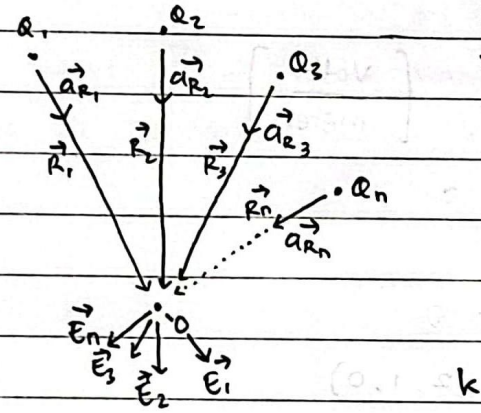
$= -3,24\ a_x + 79,38\ \vec{a_y}$ (V/m)

$|\vec{E_t}| = \sqrt{(-3,24)^2 + (79,38)^2} = 79,45$ Volt/meter

↳ intensitas medan listrik yang dirasakan di titik observator.

↳ jika ada 2/lebih medan listrik maka intensitas akan lebih kuat.

Video 10 :



$\vec{E}$ di observator O :

$\vec{E}_{total} = \vec{E_1} + \vec{E_2} + \vec{E_3} + \dots + \vec{E_n}$

$= k \frac{Q_1}{R_1^2} \vec{a_{R_1}} + k \frac{Q_2}{R_2^2} \vec{a_{R_2}} + \dots + k \frac{Q_n}{R_n^2} \vec{a_{R_n}}$

$\vec{E}_{total} = k \sum_{N=1}^{N=n} \frac{Q_N}{R_N^2} \vec{a_{R_N}}$

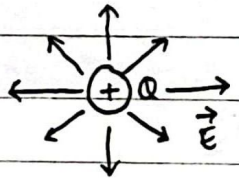
$k = \frac{1}{4\pi \varepsilon_0 \varepsilon_r}$; $\varepsilon_0 = \frac{1}{36\pi} \cdot 10^{-9}$ [Permitivitas ruang hampa]

$\varepsilon_r \geq 1$ [Permitivitas relatif medium]

$k = \frac{9.10^9}{\varepsilon_r}$ → konstanta medium

Video 11 :

Fluks Listrik (Ψ Psi)



⟶ = 1 C

1 garis medan $\vec{E}$ (fluks listrik)

$\Psi \approx Q+$

Jika $Q = 1000$ C → $\Psi = 1000$ C [1000 garis medan listrik]

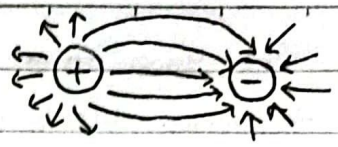
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$Q+$: medan listrik dihasilkan $Q+$

$Q-$: medan listrik menerima dr positive

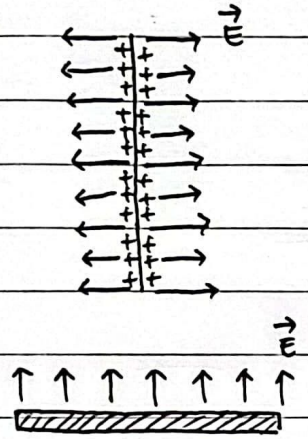


fluks searah dengan medan listrik.

Video 12 :

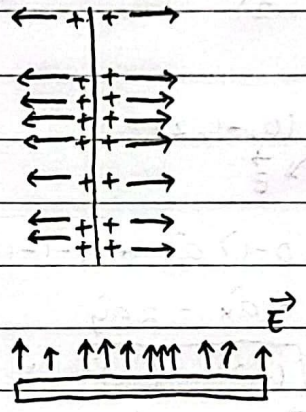
Fluks Listrik dari Muatan Listrik Terdistribusi

Distribusi Uniform



jarak antar garis fluks sama

Distribusi Non Uniform



jarak garis fluks tidak sama

fluks listrik distribusinya tergantung sejauh mana medra yg dialiri dapat mengkonsentrasikan sehingga sampai ke suatu tempat.

χ_e : Suseptibilitas elektrik

$\chi_e = \varepsilon_r - 1$

- udara (ruang bebas) [free space]
  Lp $\varepsilon_r = 1$
- air laut $\varepsilon_r > 1$

$\vec{B}$ = kerapatan fluks listrik $\left[\frac{C}{m^2}\right]$

$\vec{D} = \varepsilon_0 \varepsilon_r \vec{E}$

Medan listrik yang besar akan memperbesar kerapatan fluks listrik

[$\vec{D}$ sejajar dengan $\vec{E}$]

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Review Materi

① Apa yang kalian pahami tentang medan listrik ?

→ Medan listrik adalah daerah pengaruh dari sumber listrik (Q+).

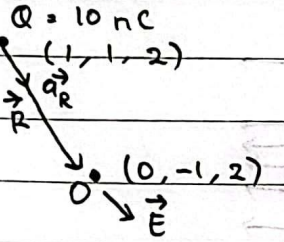
② Apa sumber medan listrik ?

→ Muatan positif (Q+)

③ Bagaimana karakteristik medan listrik ?

→ Intensitas medan listrik dirasakan di titik observator, jika ada 2 atau lebih medan listrik maka intensitas akan lebih kuat.

④ Tuliskan contoh soal tentang medan listrik !

→ 

$\vec{E} = k \cdot \frac{Q}{R^2} \vec{a}_R$

$\vec{R}_{PO} = (0-1)\hat{a}_x + (-1-1)\hat{a}_y + (2-2)\hat{a}_z$

$= -\hat{a}_x - 2\hat{a}_y$

$|\vec{R}_{PO}| = \sqrt{(-1)^2 + (-2)^2} = \sqrt{5}$

$\vec{a}_R = \frac{\vec{R}_{PO}}{|\vec{R}_{PO}|} = \frac{-\hat{a}_x - 2\hat{a}_y}{\sqrt{5}} = -0,447\,\hat{a}_x - 0,894\,\hat{a}_y$

$\vec{E} = 9 \cdot 10^9 \cdot \frac{10 \cdot 10^{-9}}{(\sqrt{5})^2} (-0,45\,\hat{a}_x - 0,89\,\hat{a}_y)$

$\vec{E} = \frac{90}{(\sqrt{5})^2} (-0,45\,\hat{a}_x - 0,89\,\hat{a}_y)$

$\vec{E} = -8,1\,\hat{a}_x - 16,02\,\hat{a}_y$ V/m

$|\vec{E}| = \sqrt{(-8,1)^2 + (-16,02)^2} = 17,95$ V/m

⑤ Apa yang kalian pahami tentang fluks listrik ?

→ fluks listrik adalah garis medan listrik, fluks searah dengan medan listrik.

⑥ Apa hubungan medan listrik dengan kerapatan fluks listrik ?

→ Medan listrik yang besar akan memperbesar kerapatan fluks listrik.

$\vec{D} = \varepsilon_0 \varepsilon_r \vec{E}$

$\vec{D} \uparrow$ maka $\vec{E} \uparrow$; berbanding sejajar.

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(7) Tuliskan contoh soal tentang hubungan medan listrik dengan kerapatan fluks listrik !

→ Diketahui : $Q = 10\,nC$ pada titik P (1, 1, 2) dan titik Oobservator pada koordinat (0, −1, 2). Jika kedua titik berada pada dielektrik air laut, maka tentukan:

a) $\vec{E}$

b) $|\vec{E}|$

c) $\vec{D}$

→ Jawab : $\vec{R}_{PO} = (0-1)\hat{a}_x + (-1-1)\hat{a}_y + (2-2)\hat{a}_z = -\hat{a}_x - 2\hat{a}_y$

$|\vec{R}_{PO}| = \sqrt{5}$

$\vec{a}_{RPO} = \dfrac{\vec{R}_{PO}}{|\vec{R}_{PO}|} = -0,447\,\hat{a}_x - 0,894\,\hat{a}_y$

a) $\vec{E} = \dfrac{9.10^9}{2,2} \cdot \dfrac{10.10^{-9}}{(\sqrt{5})^2} (-0,45\,\hat{a}_x - 0,89\,\hat{a}_y)$

$= \dfrac{90}{2,2.5} (-0,45\,\hat{a}_x - 0,89\,\hat{a}_y)$

$= 8,18\ (-0,45\,\hat{a}_x - 0,89\,\hat{a}_y)$

$= -3,681\,\hat{a}_x - 7,28\,\hat{a}_y$ V/m

b) $|\vec{E}| = \sqrt{(-3,68)^2 + (-7,28)^2} = 8,15$ V/m

c) $\vec{D} = \varepsilon_0\,\varepsilon_r\,\vec{E}$

$\vec{D} = \frac{1}{36\pi} \cdot 10^{-9} \cdot 2,2 \cdot (-3,681\,\hat{a}_x - 7,28\,\hat{a}_y)$

$\vec{D} = 19,45 \cdot 10^{-12} (-3,681\,\hat{a}_x - 7,28\,\hat{a}_y)$

$\vec{D} = -71,6\ldots \cdot 10^{-12}\,\hat{a}_x - 141,59 \cdot 10^{-12}\,\hat{a}_y$ C/m² //.

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